

Friction Compensation Based on the LuGre Friction Model

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Abstract—This article proposes two types of simple disturbance observers for friction compensation. The proposed observers are based on the LuGre friction model. One observer directly uses the LuGre model as a design base, where the LuGre model is locally linearized and observer gains are derived from the linearized model. The other one is based on a type-1 disturbance observer with the LuGre model. In the later case, the LuGre model is used in a feedforward manner. Simple simulation to evaluate positioning accuracy of observer-based state feedback control systems were conducted, and the results showed that both observers worked well for cancelling friction force in the presence of plant parameter perturbations.

Keywords: *LuGre friction model, disturbance observer, friction compensation.*

I. INTRODUCTION

For nanoscale servo systems, friction force/torque is one of the most critical obstacles against achieving required positioning accuracy. Friction compensation using type-1 disturbance observers (DOBs) or other kinds of DOBs such as in [1] is a common way to solve this issue.

The LuGre friction model [2]–[5] is a general friction model and has been recently widely used for a basis of friction compensation [6]–[11]. There also have been parameter identification methods proposed [12][13]. Thus, these research tendencies have accelerated the use of the LuGre model in practical control problem.

This article proposes two disturbance observers based on the LuGre model and show the validity by simulation.

II. LU GRE FRICTION MODEL

According the LuGre model [2]–[5], the parameters and variables are listed in Table I, friction force is given by

$$F_L = \sigma_0 z + \sigma_1 \dot{z} + \sigma_2 v \quad (1)$$

where dynamics of deformation of bristles z , *i.e.*, an internal state, are described as follows:

$$\frac{d}{dt} z = v - \frac{\sigma_0 |v|}{g(v)} z, \quad (2)$$

TABLE I. PHYSICAL PARAMETERS AND VARIABLES

Description	Symbol	Value [5]
Coulomb friction force	F_C	2.94 N
Static friction force	F_S	5.88 N
Bristle stiffness	σ_0	2940 N/m
Bristle damping	σ_1	108 kg/s
Viscous friction coefficient	σ_2	0 kg/s
Stribeck velocity	v_S	0.001 m/s
Mass of the system	m	1 kg
Position of the system	p	N/A
Velocity of the system	v	N/A
Bristle deformation	z	N/A

$$g(v) = F_C + (F_S - F_C) e^{-\left| \frac{v}{v_S} \right|}. \quad (3)$$

In the following part, the term of viscous friction $\sigma_2 v$ is considered to be zero because it is linear and easy to be compensated, and the friction force here is assumed to be represented as

$$F = \sigma_0 z + \sigma_1 \dot{z} \quad (4)$$

although $\sigma_2 v$ is incorporated with the state equation below.

Using the LuGre model, one mass system under the affection of friction is described as

$$\frac{d}{dt} \begin{bmatrix} p \\ v \\ z \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -\frac{\sigma_1 + \sigma_2}{m} & -\frac{\sigma_0 - \sigma_0 \alpha(v)}{m} \\ 0 & 1 & -\sigma_0 \alpha(v) \end{bmatrix} \begin{bmatrix} p \\ v \\ z \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ m \end{bmatrix} u \quad (5)$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} p \\ v \\ z \end{bmatrix} \quad (6)$$

where

$$\alpha(v) = \frac{|v|}{g(v)}. \quad (7)$$

For the system above, the friction force is given by

$$F = \begin{bmatrix} 0 & \sigma_1 & \sigma_0 - \sigma_0 \sigma_1 \alpha(v) \end{bmatrix} \begin{bmatrix} p \\ v \\ z \end{bmatrix}. \quad (8)$$

III. FRICTION OBSERVER

A. LuGre Model Based Disturbance Observer

First, a nonlinear disturbance observer is designed. The observer gain $K(\alpha_m(v)) = [k_1(\alpha_m(v)) \ k_2(\alpha_m(v)) \ k_3(\alpha_m(v))]^T$ is derived by pole placement assuming that the nonlinear term $\alpha_m(v)$ can be replaced by a constant at that time.

The designed observer is given by

$$\frac{d}{dt} \begin{bmatrix} \hat{p} \\ \hat{v} \\ \hat{z} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -\frac{\sigma_{1m} + \sigma_{2m}}{m} & -\frac{\sigma_{0m} - \sigma_{0m} \sigma_{1m} \alpha_m(v)}{m} \\ 0 & 1 & -\sigma_{0m} \alpha_m(v) \end{bmatrix} \begin{bmatrix} \hat{p} \\ \hat{v} \\ \hat{z} \end{bmatrix}, \quad (9)$$

$$+ \begin{bmatrix} 0 \\ \frac{1}{m} \\ 0 \end{bmatrix} u + \begin{bmatrix} k_1(\alpha_m(v)) \\ k_2(\alpha_m(v)) \\ k_3(\alpha_m(v)) \end{bmatrix} (p - \hat{p})$$

$$\begin{bmatrix} \hat{p} \\ \hat{v} \\ \hat{F} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & \sigma_{1m} & \sigma_{0m} - \sigma_{0m} \sigma_{1m} \alpha_m(v) \end{bmatrix} \begin{bmatrix} \hat{p} \\ \hat{v} \\ \hat{z} \end{bmatrix} \quad (10)$$

where the subscript m show that the parameters are identified ones. Note that the modeled nonlinear term, $\alpha_m(v)$, is used in Equations (9) and (10). Although the stability analysis should be carefully discussed, by using the $\alpha_m(v)$, feedforward effect can be utilized. Our future work would be to improve this stability issue by using state-dependant Riccati equation (SDRE) method. Figure 1 shows an example of a state-feedback position control system based on the observer proposed in this section.

B. Type-1 Disturbance Observer with LuGre Model

In the second one, the LuGre model is used only as a feedforward compensator to generate a friction estimate $\hat{F}$. When modeling errors due to controlled plant uncertainties exist, a type-1 DOB makes an adjustment for friction by the

term $d\hat{F}$. This type-1 disturbance observer with LuGre model is here named “dual observer.” The dual observer is given by

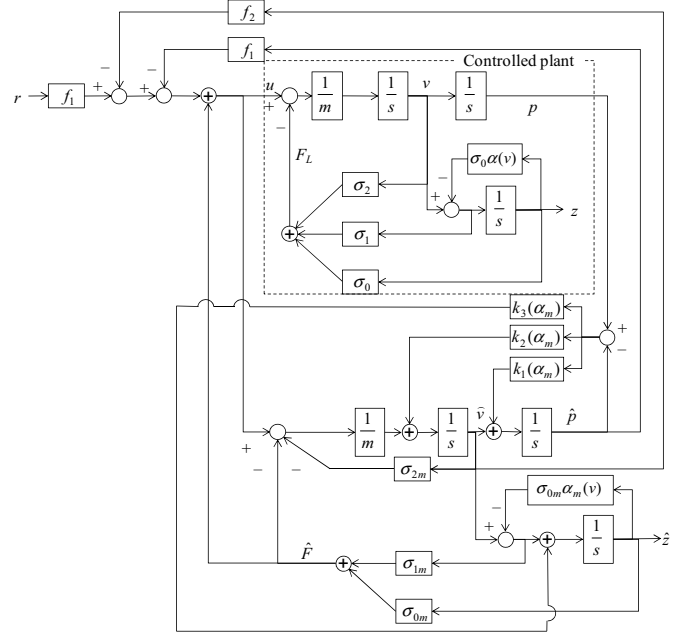


Figure 1. Position control system using LuGre based DOB

$$\begin{aligned} \frac{d}{dt} \begin{bmatrix} \hat{p} \\ \hat{v} \\ d\hat{F} \\ \hat{z} \end{bmatrix} &= \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -\frac{\sigma_{2m}}{m} & -\frac{1}{m} & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & -\sigma_{0m} \alpha_m(v) \end{bmatrix} \begin{bmatrix} \hat{p} \\ \hat{v} \\ d\hat{F} \\ \hat{z} \end{bmatrix} \\ &+ \begin{bmatrix} 0 \\ \frac{1}{m} \\ 0 \\ 0 \end{bmatrix} u' + \begin{bmatrix} k_1 \\ k_2 \\ k_3 \\ 0 \end{bmatrix} (p - \hat{p}) \\ &= \begin{bmatrix} & 0 & & \\ A_{T1} & 0 & & \\ & 0 & & \\ 0 & 1 & 0 & -\sigma_{0m} \alpha_m(v) \end{bmatrix} \begin{bmatrix} \hat{p} \\ \hat{v} \\ d\hat{F} \\ \hat{z} \end{bmatrix} \\ &+ \begin{bmatrix} B_{T1} \\ 0 \end{bmatrix} u' + \begin{bmatrix} K \\ 0 \end{bmatrix} (p - \hat{p}) \end{aligned} \quad (11)$$

$$\begin{bmatrix} \hat{p} \\ \hat{v} \\ d\hat{F} \\ \hat{F} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & \sigma_{1m} & 0 & \sigma_{0m} - \sigma_{0m}\sigma_{1m}\alpha_m(v) \end{bmatrix} \begin{bmatrix} \hat{p} \\ \hat{v} \\ d\hat{F} \\ \hat{z} \end{bmatrix} \quad (12)$$

$$= \begin{bmatrix} C_{T1} & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & \sigma_{1m} & 0 & \sigma_{0m} - \sigma_{0m}\sigma_{1m}\alpha_m(v) \end{bmatrix} \begin{bmatrix} \hat{p} \\ \hat{v} \\ d\hat{F} \\ \hat{z} \end{bmatrix}$$

where the observer gain $K = [k_1 \ k_2 \ k_3]^T$ can be determined for the type-1 DOB part (A_{T1} , B_{T1} , C_{T1}) by pole placement or other linear control methods. Note that the adding points of $\hat{F}$ and $d\hat{F}$ is different and the observer input u' is taken from between these two points.

Figure 2 shows an example of a state-feedback position control system based on the observer proposed in this section.

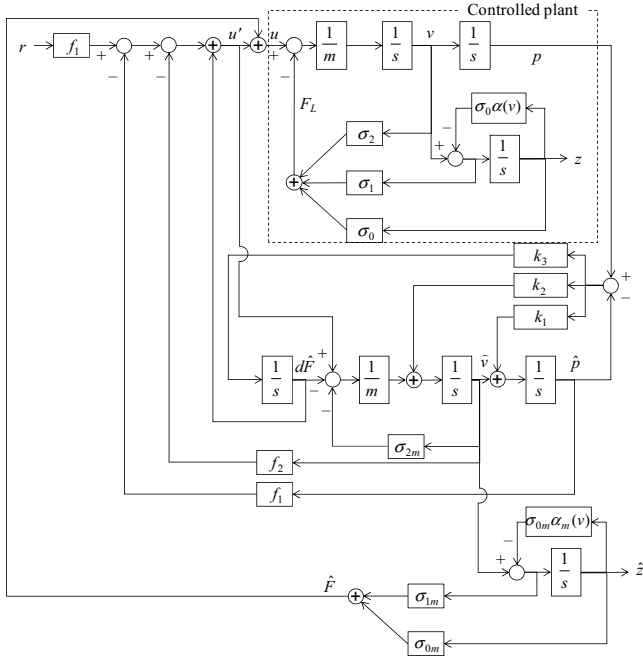


Figure 2. Position control system using dual DOB

IV. SIMULATION STUDY

A. Friction Estimation

Figures 3 and 4 shows simulation results to see friction estimates respectively using LuGre based DOB and Dual DOB when the input forces are also respectively given by $8\cos(2\pi \cdot 2t)$ and $6\cos(2\pi \cdot 2t)$. In these simulation, parameter values listed in Table I are used, cited from [5] as a benchmark. Observers' poles in the simulation are listed in Table II. The plant parameters, F_C , σ_0 and σ_1 , are perturbed by +20% in the simulation although observer design and implementation were done with the nominal values. For both simulations, friction estimates are fed back to the controlled plant to compensate the friction. Thus, the position should not be drifted for the input of cosine waveforms, but the position in Figure 3 is slightly shifted. On the other hand, there is no drift in the dual DOB case.

B. Position Control

To show the validity of the proposed method, responses to a step input of 2 mm were evaluated. As shown in Figures 1 and 2, the positioning systems used here are observed-state feedback control systems, the poles of which are listed in Table II. Figures 5 and 6 show simulation results, i.e., responses of the positioning systems respectively in Figures 1 and 2 for a step input of 2 mm in the presence of the same plant perturbations as described in Section IV A. As shown in these plots, both DOBs work well to cancel the affection of frictions and to eliminate position offset due to the friction.

TABLE II. OBSERVER AND CONTROLLER PARAMETERS

Description	LuGre Model Based DOB	Type-1 DOB with LuGre Model
Observer poles	$-2\pi \cdot 30, 2\pi \cdot 30 \cdot (-2 \pm \sqrt{2^2 - 1})$ [rad/s]	$-2\pi \cdot 30, 2\pi \cdot 30 \cdot (-2 \pm \sqrt{2^2 - 1})$ [rad/s]
Gain k_1	Function of α_m	9.4248×10^2
Gain k_2	Function of α_m	1.7765×10^5
Gain k_3	Function of α_m	-6.6974×10^6
State feedback poles	$2\pi \cdot 10 \cdot (-0.7 \pm \sqrt{1 - 0.7^2} \cdot j)$ [rad/s]	$2\pi \cdot 10 \cdot (-0.7 \pm \sqrt{1 - 0.7^2} \cdot j)$ [rad/s]
Gain f_1	3.9478×10^3	3.9478×10^3
Gain f_2	8.7965×10^1	8.7965×10^1

V. CONCLUSION

This article proposed two types of disturbance observers based on the LuGre friction model. One observer is based on a locally-linearize LuGre model and has time varying observer gain as a function of $\alpha_m(v)$. The other one is based on a type-1 disturbance observer with the LuGre model, which is used in a feedforward manner. Simulation results showed both DOBs work well to cancel the affection of frictions and to eliminate position offset due to the friction in the presence of plant perturbations.

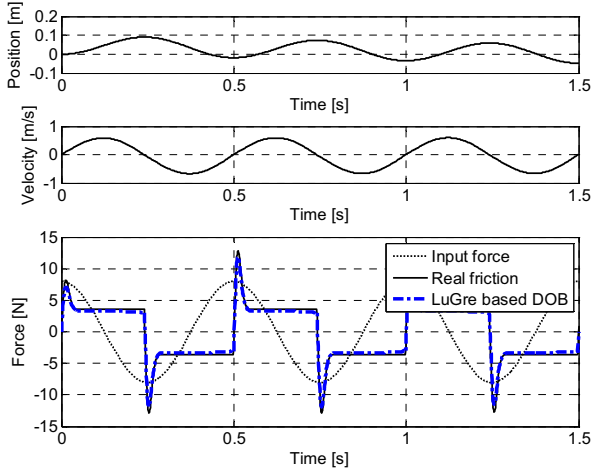


Figure 3. Friction estimation results of LuGre based DOB for a cosine input with plant parameter perturbations.

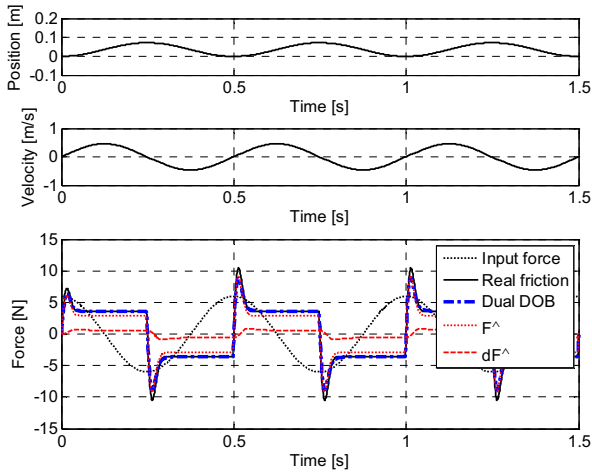


Figure 4. Friction estimation results of dual DOB for a cosine input with plant parameter perturbations accompanied with $\hat{F}$ and $d\hat{F}$.

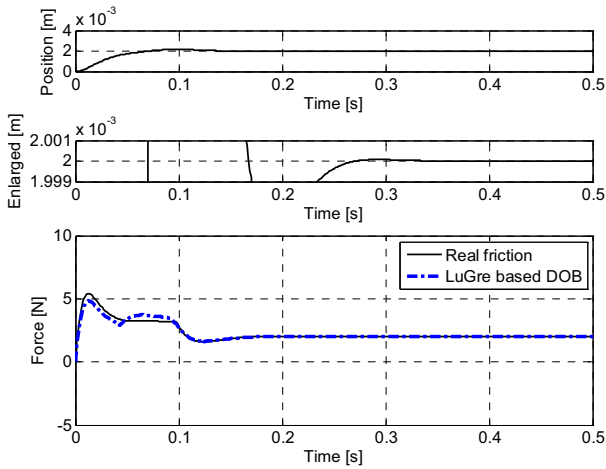


Figure 5. Step response of 2 mm with LuGre based DOB.

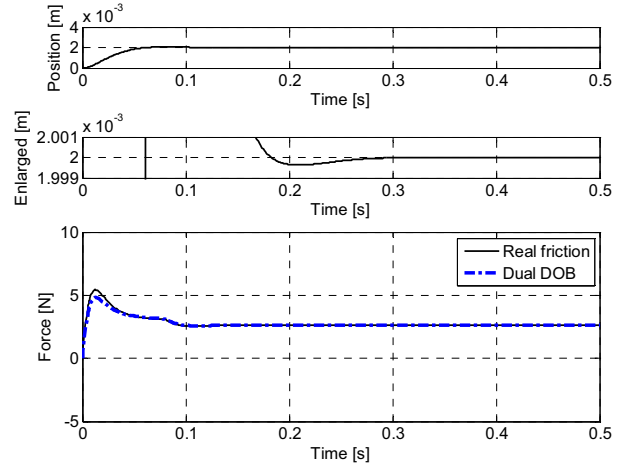


Figure 6. Step response of 2 mm with dual DOB.

REFERENCES

- [1] J. Ishikawa, M. Tomizuka, "Pivot Friction Compensation Using an Accelerometer and a Disturbance Observer for Hard Disk Drives," *IEEE/ASME Trans. on Mechatronics*, vol. 3, no. 3, pp. 194-201, 1998.
- [2] C. Canudas de Wit, H. Olsson, K.J. Åström, and P. Lischinsky, "A new model for control of systems with friction," *IEEE Trans. on Automatic Control*, vol. 40, no. 3, pp. 419-425, Mar. 1995.
- [3] C. Canudas de Wit, H. Olsson, K.J. Åström, and P. Lischinsky, "Dynamic friction models and control design," *Proc. of the 1993 American Control Conference*, pp. 1920-1926, 1993.
- [4] H. Olsson, K. J. Åström, C. Canudas de Wit, M. Gäfvert, and P. Lischinsky, "Friction models and friction compensation," *Euro. Journal of Control*, vol. 3, pp. 176-195, 1998.
- [5] K.J. Åström, C. Canudas, "Revisiting the LuGre friction model," *IEEE Control Systems Magazine*, Vol. 28, Issue 6, pp. 101 - 114, 2008.
- [6] A. Shiriaev, A. Robertsson, and R. Johansson, "Friction compensation for passive systems based on the LuGre model," *Proc. of the 2nd IFAC Workshop on Lagrangian Hamiltonian Methods Nonlinear Control*, pp. 183 - 188, 2003.
- [7] A. Robertsson, A. Shiriaev, and R. Johansson, "Friction compensation for nonlinear systems based on the LuGre model," *Proc. of the 6th Symposium on Nonlinear Control Systems*, pp. 1439-1444, 2004.
- [8] Y. Wang, Z. Xiong, H. Ding, X. Zhu, "Nonlinear friction compensation and disturbance observer for a high-speed motion platform," *Proc of the 2004 IEEE International Conference on Robotics and Automation (ICRA '04)*, Vol. 5, pp. 4515 - 4520, 2004.
- [9] T. Dumitriu, M. Culea, T. Munteanu, E. Ceanga, "Friction Compensation for Accurate Positioning in DC Drive Tracking System," *Proc. of the 2006 3rd International Conference on Electrical and Electronics Engineering*, pp. 1 - 4, 2006.
- [10] L. Lu, B. Yao, Q. Wang, Z. Chen, "Adaptive robust control of linear motor systems with dynamic friction compensation using modified LuGre Model," *Proc. of the IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM 2008)*, pp. 961 - 966, 2008.
- [11] L. Freidovich, A. Robertsson, A. Shiriaev, R. Johansson, "LuGre-Model-Based Friction Compensation," *IEEE Transactions on Control Systems Technology*, Vol. 18, Issue 1, pp. 194 - 200, 2010.
- [12] J. Han, Y. Sun, Y. Wang, "Parameter Identification of LuGre Tire Model for the Simplified Motion Dynamics of a Quarter-vehicle Model Based on Ant Colony Algorithm," *Proc. of the 2007 IEEE International Conference on Automation and Logistics*, pp. 3039 - 3043, 2007.
- [13] Z. Wenjing, "Parameter Identification of LuGre Friction Model in Servo System Based on Improved Particle Swarm Optimization Algorithm," *Proc of the 2007 Chinese Control Conference (CCC 2007)*, pp. 135 - 139, 2007.